1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist synthesis, TIL-rich tumors (or tumors with high inferred antigenicity from mismatch-repair defects) do not always enjoy prolonged clinical control — why does high immune presence sometimes fail to translate into delayed progression? 2) Tier 1 paths used: I built on the MSH6 (mismatch-repair → increased neoantigen load → expected higher TILs) path and the MDK (tumor-secreted growth factor carried in EVs → immune-modulatory, pro-tumorigenic signaling) path because they give a clear, testable antagonistic mechanism (increased antigenicity versus tumor-driven immune suppression). 3) Tier 2 evidence supporting the claim: KRT17/EV signatures (supporting an EV-mediated tumor program), immunotherapy-biomarker literature and lymph-node metastasis (supporting immune/tumor-interaction relevance), and PTGDS/HDL-C (as systemic lipid/inflammatory modulators that could tune immune function) all reinforce a model where tumor-intrinsic programs modulate whether TILs are effective. 4) Why PFI as endpoint: Progression-free interval is the most directly relevant clinical readout for whether intratumoral immune activity controls tumor growth over time (more sensitive than OS to early immune-evasion mechanisms) and is available in TCGA-style datasets. 5) Deliberate exclusions: I excluded TLS-like aggregates and TSR from the core hypothesis to keep the mechanistic focus on immune cell abundance vs functional suppression; KRT17 and EV modalities were relegated to secondary_modifiers to avoid exceeding the 2-gene anchor limit. I excluded distal/methodological paths (CSF-AD, AI accessibility, PSA cross-context) from the core claim because they are either weakly mechanistic for this puzzle or implementation-focused rather than explanatory?

Working Hypothesis

In TCGA-BRCA tumors, high tumor-infiltrating lymphocyte (TIL) density predicts longer progression-free intervals (PFI) in MSH6-proficient tumors, reflecting higher neoantigen load and effective anti-tumor immunity. However, in tumors with MSH6 deficiency (impaired mismatch repair and elevated neoantigen burden), elevated MDK expression—reflecting MDK carried by tumor-derived extracellular vesicles (EVs)—diminishes TIL effector function and/or retention, such that high TIL density does not translate into improved PFI and may be associated with shorter PFI. This interaction is mechanistically plausible because MSH6 deficiency increases antigenicity (potentially enhancing TIL recruitment), while MDK/EV signaling suppresses TIL activity via immune checkpoints, T-cell exhaustion pathways, or suppression of antigen presentation in the tumor microenvironment. KRT17, PTGDS, and HDL-C can act as secondary modifiers that modulate the magnitude of MDK-driven suppression or the systemic/inflammatory milieu, but the primary prognostic signal arises from the MSH6–MDK–TIL axis. The specific, testable predictions are: (1) a positive association between TIL density and PFI in MSH6-proficient tumors, (2) attenuation or reversal of this benefit in MSH6-deficient tumors with high MDK expression, (3) stronger adverse impact of MDK/EV signaling in the context of high TIL density, and (4) partial restoration of TIL functionality and PFI with MDK inhibition or disruption of EV-MDK signaling. These relationships can be evaluated in BRCA data with PFI endpoints and, where feasible, validated in mechanistic models.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, MDK
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6 (mismatch-repair → increased neoantigen load → expected higher TILs) path and the MDK (tumor-secreted growth factor carried in EVs → immune-modulatory, pro-tumorigenic signaling) path because they give a clear, testable antagonistic mechanism (increased antigenicity versus tumor-driven immune suppression).

MDK: MDK (tumor-secreted growth factor carried in EVs → immune-modulatory, pro-tumorigenic signaling) path because they give a clear, testable antagonistic mechanism (increased antigenicity versus tumor-driven immune suppression).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	KRT17	Supporting modifier from knowledge graph
Secondary	PTGDS	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	lymph_node_metastasis	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In silico analysis of BRCA cohorts to test the MSH6–MDK–TIL interaction on PFI using Cox models with interaction terms, adjusting for confounders (age, stage, receptor status).

Prediction 2: Stratified analyses by MSH6 status (mutant/low expression vs. proficient/high) and by MDK expression or EV-signature scores; assess whether high TIL density confers PFI benefit only in MSH6-proficient/MDK-low groups.

Prediction 3: Functional assays in vitro: manipulate MSH6 and MDK in BRCA cell lines; co-culture with T cells; measure T-cell activation, proliferation, and cytotoxicity in the presence vs. absence of MDK inhibitors or EV-secretion blockers.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- KRT17: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PTGDS: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.